Pulsed Sensor Interference on 5G NR: Deterministic Collision Modeling and Over-the-Air Validation with OCUDU

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***Abstract*—Pulsed sensor interference poses a growing challenge to 5G New Radio (NR) deployments in shared spectrum bands, including the 3.5 GHz Citizens Broadband Radio Service (CBRS) and projected FR3 allocations for 6G. While Integrated Sensing and Comms (ISAC) efforts are underway and Spectrum Access Services (SAS) exist for CBRS, understanding these system failure effects can lead to stronger mitigation and coexistence. Prior work characterizes radar interference through duty cycle, an abstraction that conflates pulse duration and pulse repetition frequency (PRF) while obscuring their distinct effects on 5G MAC-layer performance. Existing detection approaches require IQ-level waveform analysis and dedicated sensing hardware. This paper presents a closed-form deterministic model predicting MAC-Layer PDSCH/PDCCH collision probability to System-Level per-slot downlink throughput loss as a function of PRF, pulse duration, and TDD slot pattern. Conclusions are derived entirely from standard gNB MAC-layer telemetry (throughput, BLER, MCS, and HARQ statistics) without additional spectrum sensing hardware. The model is then validated using our OCUDU/open5GS testbed with a commercial UE across 437 runs at PRF: 100 Hz, Pulse-Duration:100 microseconds, which confirms our lower-bounded model predictions to within 1.2 percentage points of measured average downlink loss. Our experimental findings also highlight the effectiveness of low cost SDRs to validate model performance, and generate large datasets required for AI-RAN efforts. Designed to extend naturally into an O-RAN xApp via MAC Layer KPIs, experimental verification showcases a strong forward mapping for future leveraging of Bayesian inverse inference to reverse identify sensor PRF and pulse duration from live network telemetry, enabling passive sensor characterization without dedicated spectrum monitoring hardware.**

***Index Terms*—5G NR, Pulsed Interference, Spectrum Coexistence, PDSCH, PDCCH, CORESET, TDD, PRF, HARQ, Throughput Degradation, Collision Modeling, O-RAN, OCUDU, CBRS, Bayesian Inference**

I. Introduction

Spectrum coexistence between commercial telecommunications, e.g., 5G New Radio (NR), and active sensing systems has moved from a planning concern to an operational reality across licensed, unlicensed, and emerging allocations such as FR3 in 6G. Currently in the 3.5 GHz Citizens Broadband Radio Service (CBRS) band (3550–3700 MHz) [1], [2], 5G deployments must already share spectrum with incumbent naval radar systems whose pulsed transmissions can periodically overlap active downlink communications, degrading throughput and disrupting operations without mitigations such as SAS (Spectrum Access System). The coexistence challenge is set to intensify with regulators planning shared-use allocations in the FR3 mid-band (7–24 GHz) for 6G [3], Integrated Sensing and Communication (ISAC) architectures under active development, and with 5G NR-U already available in unlicensed 5 GHz and 6 GHz spectrum [4], [5].

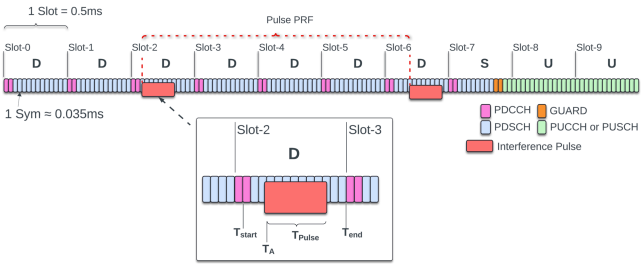


Fig. 1: Deterministic temporal overlap between pulsed interference and a 7D1S2U 5G NR TDD frame structure.

In this paper, the term pulsed sensor refers broadly to radar-like active RF sensing systems that periodically emit short-duration pulses for detection, ranging, or environmental sensing. Pulsed interference is predictable in 5G NR when two conditions hold simultaneously: the pulse carries sufficient power to degrade a received signal block, and the interference must temporally overlap with the 5G signal in a predictable manner. As illustrated in Fig. 1, periodic pulsed interference can repeatedly align with specific downlink transmission intervals within the 5G NR TDD structure, producing deterministic temporal collision patterns. Under these conditions, affected MAC-layer transport blocks experience deterministic collision-induced corruption governed by the pulse repetition frequency (PRF) and pulse duration. This

paper fixes the transmit power above the decoding disruption threshold and develops a deterministic temporal model that predicts system-level downlink throughput degradation as a function of radar PRF, pulse duration, and the 5G TDD slot structure with HARQ disabled.

A validated forward model lets network operators and spectrum managers predict 5G throughput degradation under arbitrary sensor configurations without needing live radar access, which matters for coexistence planning, link budgeting, and scheduler design. With accurate predictions, this forward mapping unlocks a more useful inverse problem: given anomalous MAC-layer KPIs (elevated BLER, dropped MCS, HARQ NACK spikes), we can infer the most likely sensor PRF and pulse duration. Bayesian inversion is a practical approach here because the forward model is closed-form, well-validated, and operates entirely on standard gNB telemetry, enabling direct integration into O-RAN xApp architectures. MAC-layer telemetry has also proven useful in other passive inference contexts, such as our previous work on indoor localization [6], reinforcing its value as a general-purpose observability layer.

This paper makes the following contributions.

- A closed-form deterministic model predicting (Downlink) PDSCH/PDCCH collision probability and throughput loss as a function of radar PRF, pulse duration, and 5G TDD slot pattern (Section III).
- Over-the-air (OTA) experimental validation using an OCUDU [7] / open5GS testbed with a commercial UE, confirming the model predictions across a range of PRF values, and independent slot collisions. (Section IV).
- A characterization of the forward model (radar parameters $\rightarrow$ MAC-layer KPI degradation) as the empirical foundation for future Bayesian inverse inference of radar parameters from standard gNB telemetry logs.

II. Related Work

Prior radar and 5G coexistence efforts primarily focused on protecting incumbent radar systems *from* commercial wireless interference, with the NTIA introducing exclusion zones and characterizing interference largely through duty cycle abstractions [8]. While duty cycle,$D = \text{PD} \times \text{PRF}$, is a useful high-level radar metric, it conflates pulse duration (PD) and pulse repetition frequency (PRF), two parameters that produce fundamentally different temporal interference behaviors in 5G NR systems.[9]. Prior coexistence studies further characterized interference primarily through radar power and duty cycle, showing that throughput degradation increases with interference power while pulse width alone has more limited observable impact [10], [11], [12]. A complementary line of work detects incumbent radar activity from IQ samples or spectrum snapshots using ESC sensors and O-RAN-integrated classifiers [13], [14], [15].

In contrast, this work treats PRF and pulse duration as independent pulsed-interference parameters and develops a closed-form deterministic framework describing how periodic pulsed interference interacts with 5G NR TDD slot structures to produce predictable MAC-Layer PDSCH/PDCCH collisions to system-level throughput degradation. Unlike IQ-sample-based sensing approaches, our approach focuses on standard gNB MAC-layer telemetry, eliminating the need for dedicated sensing hardware, and enabling practical extensions into AI-RAN / O-RAN Near-RT-RIC/xApp architectures. Our prior work established an initial experimental characterization of PRF-driven coexistence behavior [9], [16]; this paper extends that foundation through a closed-form analytical model and systematic OTA validation.

III. Deterministic Temporal Collision Model

This section establishes a general deterministic model for predicting 5G throughput degradation caused by radar pulse interference. The analysis is broken down into two parts. First is an estimation of the PDSCH collision window based on 5G and Radar Parameters. Secondly is a throughput loss based on expected block sizes without system responses of HARQ Re-Transmission, and Link Adaptation. Assumptions are outlined in Table I.

TABLE I: Model assumptions collision framework.

Type	Assumption
PHY	Any pulse-symbol overlap causes CRC failure; FEC recovery is not modeled.
PHY	PDCCH uses Polar coding and PDSCH uses LDPC; coding differences are not explicitly modeled, and considered equally fragile to pulses.
PHY	Cyclic prefix modeling is ignored due to negligible timing impact. CP « Symbol « Pulse
MAC	Semi-persistent scheduling (SPS) is assumed.
MAC	Corrupted PDCCH grants are treated equivalently to lost transport blocks.
MAC	Link adaptation is disabled / MCS remains fixed.
MAC	HARQ retransmissions are disabled and not modeled.
MAC	The system is fully loaded with sufficient traffic to utilize all downlink symbols.
Scope	SSB/MIB/SIB reduce slot-0 PDSCH allocation.
Scope	Pulse duration satisfies $0 < t_{\text{pulse}} \leq t_{\text{PDSCH-D}}$.

TABLE II: Model Notation descriptions.

Variable	Description	Variable	Description
T_A	Pulse start time	t_{pulse}	Radar pulse duration
$T_{\text{start},k}$	Start time of slot k	t_{PDCCH}	PDCCH duration
$T_{\text{end},k}$	End time of slot k	$t_{\text{PDSCH-D}}$	PDSCH dur. for full DL slots
t_{slot}	Slot duration	$t_{\text{PDSCH-S}}$	PDSCH dur. for special slots
N_D	Num DL slots / pattern	N_{Total}	N Total slots / TDD pattern

A. Temporal Collision Probability Model

The proposed deterministic framework models how periodic pulsed interference temporally overlaps active 5G NR downlink transmission regions within a fixed TDD slot structure. The model consists of two coupled components: (i) a temporal collision probability model that characterizes overlap between interference pulses and PDSCH/PDCCH

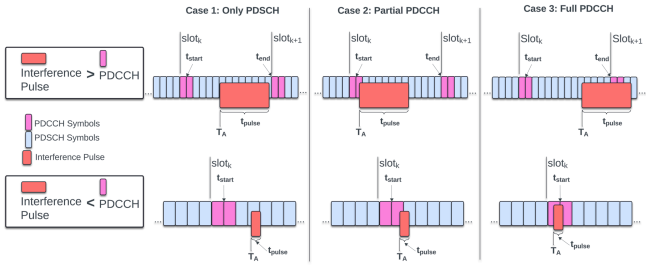


Fig. 2: Downlink collision cases: PDSCH-only (Case 1), PDCCH boundary (Case 2), and PDCCH (Case 3).

TABLE III: Collision probabilities for 7D1S2U ($t_{\text{PDCCH}} = 71.4\,\mu\text{s}$, $t_{\text{slot}} = 500\,\mu\text{s}$).

Case	$t_p = 100\,\mu\text{s}$	$t_p = 20\,\mu\text{s}$
P_1 (PDSCH only)	53.1%	64.3%
P_2 (PDCCH boundary)	20.0%	5.6%
P_3 (PDCCH multi-slot / interior)	4.0%	7.2%
Total	77.1%	77.1%

transmission intervals using notation summarized in Table II. And (ii) a throughput degradation model that maps collision events to expected throughput loss using transport block size (TBS) analysis, and values in Table V.

As illustrated in Fig. 2, a pulse of duration t_{pulse} starting at T_A collides with slot k if it overlaps the PDSCH region, the PDCCH region, or both (Fig. 1). When $t_{\text{pulse}} > t_{\text{PDCCH}}$ the pulse bridges the control boundary and can affect two adjacent slots; when $t_{\text{pulse}} \le t_{\text{PDCCH}}$ collisions are confined to a single region. The closed-form probabilities for N_D full DL slots and one special slot per pattern are:

$$P_1 = \frac{N_D(t_{\text{PDSCH-D}} - t_{\text{pulse}}) + t_{\text{PDSCH-S}}}{N_{\text{Total}}\, t_{\text{slot}}} \tag{1}$$

$$P_2 = \frac{2N_D\, \min(t_{\text{pulse}},\, t_{\text{PDCCH}})}{N_{\text{Total}}\, t_{\text{slot}}} \tag{2}$$

$$P_3 = \frac{N_D\, |t_{\text{pulse}} - t_{\text{PDCCH}}|}{N_{\text{Total}}\, t_{\text{slot}}} \tag{3}$$

where

- P_1 represents PDSCH-only collisions,
- P_2 represents PDCCH boundary collisions,
- P_3 represents PDCCH interior or slot-bridging collisions.

These sum to the DL symbol occupancy of the TDD pattern: $P_1 + P_2 + P_3 = N_{\text{DL sym}}/N_{\text{sym/slot}}$. For 7D1S2U this is $108/140 = 77.1\%$. Table III verifies this for two representative pulse durations, using a 2 symbol PDCCH as per Table V and shows that shorter pulses shift probability from Cases 2&3 into Case 1.

PDSCH transmissions are protected using LDPC coding, while PDCCH transmissions use Polar coding. LDPC codes are generally more sensitive to interference than Polar codes [17]. Keeping our assumptions in Table I, SPS scheduling, and HARQ retransmissions disabled, a corrupted/unscheduled PDCCH grant produces the same net result as a lost/decode-failed PDSCH, (An unscheduled TBS is equivalent to a lost Block). Therefore, both collision types are modeled as equivalent throughput degradation events.

When $t_{\text{pulse}} < t_{\text{PDCCH}}$, the expected DL throughput degradation associated with a collision in slot k is approximated as

$$T_P(k) = (P_1 + P_2 + P_3)(\text{PRF} \cdot \text{TBS}_k) \tag{4}$$

When $t_{\text{pulse}} > t_{\text{PDCCH}}$, pulses may bridge adjacent slots with probability P_3, producing coupled transport block losses across neighboring slots. The expected throughput degradation then becomes

$$T_P(k) = (P_1 + P_2)(\text{PRF} \cdot \text{TBS}_k) + P_3\,(\text{PRF} \cdot (\text{TBS}_k + \text{TBS}_{k-1})) \tag{5}$$

B. Slot-Level Throughput Degradation

The temporal collision probability model can be mapped to expected throughput degradation by associating each collision event with the TBS scheduled within the affected slot. Per-slot TBS values are determined using the 3GPP NR TBS lookup procedure [18]. For the configured 20 MHz, 30 kHz SCS, 256-QAM (MCS 27) system, slot 0 carries a reduced allocation due to SSB PRB reservation (TBS = 3521 bytes), full downlink slots 1–6 carry the maximum allocation (TBS = 5889 bytes), and special slot 7 carries a reduced downlink symbol allocation (TBS = 2817 bytes).

A PRF of 100 Hz aligns with the 10 ms periodicity of the 30 kHz SCS frame structure, producing one interference pulse opportunity per frame interval and enabling temporally repeatable collision behavior across the slot sequence as summarized in Table IV.

IV. Over-the-Air Validation Platform and Experimental Design

A. Experimental Platform & Configuration

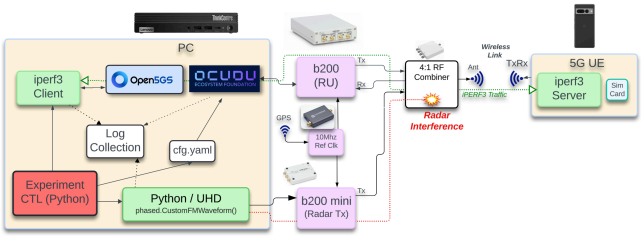


Fig. 3: OTA experimental platform for generating pulsed interference using OCUDU and open5GS.

The OTA validation platform [19], illustrated in Fig. 3, runs OCUDU [7] and open5GS on a mini-PC connected to a USRP B200 SDR, with a commercial Google Pixel 7

TABLE IV: Probabilities and Impact on DL Slots for $t_{\text{Pulse}} = 100(\mu s)$ and PRF of 100Hz.

Slot$_k$	P_{slot}	P_1, P_2/ Loss(Mbps)	P_3/ Loss(Mbps)
0	5%	4.74% / 2.82	0.26% / 2.82
1	5%	4.74% / 4.71	0.26% / 7.53
2	5%	4.74% / 4.71	0.26% / 9.42
3	5%	4.74% / 4.71	0.26% / 9.42
4	5%	4.74% / 4.71	0.26% / 9.42
5	5%	4.74% / 4.71	0.26% / 9.42
6	5%	4.74% / 4.71	0.26% / 9.42
7 S (DL)	3.57%	3.38% / 2.25	0.19% / 6.96
8	-	-	-
9	-	-	-
10	5%	4.74% / 4.71	0.26% / 4.71
11	5%	4.74% / 4.71	0.26% / 9.42
12	5%	4.74% / 4.71	0.26% / 9.42
13	5%	4.74% / 4.71	0.26% / 9.42
14	5%	4.74% / 4.71	0.26% / 9.42
15	5%	4.74% / 4.71	0.26% / 9.42
16	5%	4.74% / 4.71	0.26% / 9.42
17 S (DL)	3.57%	3.38% / 2.25	0.19% / 6.96
18	-	-	-
19	-	-	-
P_{Total}/AVG	77.1%	73.1% / 4.28	4.0% / 8.29

Pro as the UE. Pulsed radar interference was generated in Python using the UHD library and transmitted over a USRP B200 mini; both SDRs shared a 10 MHz reference clock. The combined 5G and radar signals were routed through a 4:1 RF combiner to a short-range antenna within 5 cm of the UE on a workbench.

Table V summarizes the OTA experimental parameters used throughout the evaluation. Radar interference was generated using 100 μs pulses centered at 3417.1 MHz with a PRF of 100 Hz causing approx. 5PRB overlap of the total 51PRB of the gNB. The scheduler is stressed in the UL and DL direction through iperf3 traffic, with the intent to fully stress scheduler resources.

B. Experimental Procedure and Data Collection

The automated OTA validation procedure is summarized in Fig. 4. At 100 Hz PRF, one pulse opportunity occurs per 10 ms frame, producing temporally repeatable collision patterns across the 20-slot sequence. Collision-induced failures appear in the MAC-layer logs as HARQ `ack=0` events on the PUCCH, enabling direct observation of deterministic pulse-to-slot collision behavior. Direct interference slot timing is not controlled, and considered variable. We address this through multiple runs to cover the probability of impact for each slot. The start time of a given slot can occur anywhere between 0-19 frame slots however because the PRF (100Hz) = 1 Frame (10ms), we expect that collision to remain consistent over the 20s interference duration. The full dataset of runs is publicly available at [20].

V. Model Validation And Experimental Results

Our OTA experimental results challenged assumptions made in Table I [(1)(2)] and produced four major findings. First, the forward model predicts per-slot lower-bounded throughput degradation to within 1% of measured loss, showcasing high accuracy. Second, HARQ process exhaustion near UL slots reduces PDSCH grant rates for slots 6/7 and 16/17, in a 7D1S2U pattern, independent of interference. Third, low-cost SDR performance can inflate variances in large run sets but is identifiable from telemetry. Fourth, CORESET symbol allocation governs PDCCH/PDSCH boundary geometry and is critical for accurate multi-slot collision modeling with the frequency domain.

TABLE V: OTA experimental configuration parameters.

gNB Configurations			
Band	n78	**DL ARFCN**	627340
Center Freq.	3410 MHz	**Bandwidth**	20 MHz
SCS	30 kHz	**TDD Pattern**	7D1S2U
Special Slot	10D2G2U	**Symbols/Slot**	14
Max HARQ	16	**MCS Table**	256QAM
Link Adapt.	MCS=27	**HARQ Retx**	Disabled
TX Gain	85	**RX Gain**	50
PDCCH / CORESET#0 Configuration			
CORESET	2 symbols	**CORESET PRB**	24
Agg. Level	4	**Candidates**	1 per slot
Mon Symbol	0	**PDSCH Start**	Symbol 2
DCI Format	0_0 / 1_0	**Search Space**	(SS#1)
Radar Configurations			
Pulse Dur	100 μs	**PRF**	100 Hz
Pulse BW	2 MHz	**Pulse f_c**	3417 MHz
Radar TX Gain	80	**Ref. Clock**	10 MHz
UE Configurations			
UE Type	Pixel 7 Pro	**UE Distance**	5 cm

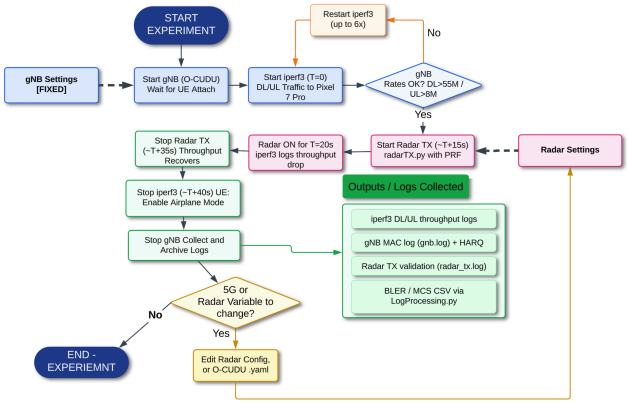


Fig. 4: Automated OTA validation procedure for pulsed interference experiments.

A. Per-Slot Throughput Degradation

Figure 5 compares model predictions against measured per-slot throughput degradation across 437 OTA runs spanning the full 20-slot 7D1S2U structure. Three distinct observations are visible: full DL slots (1–6, 10–16) consistently cluster above the model prediction, special slots (7, 17) sit lower due to reduced DL symbol count, and slot 0

is reduced by PBCH scheduling reservation. A consistent upward offset from the model floor is visible across all slot types, suggesting unmodeled details rather than random sources of additional loss.

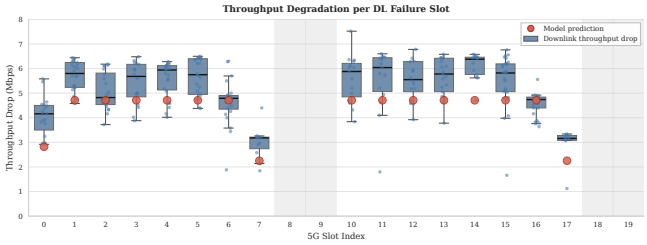


Fig. 5: IQR downlink throughput degradation for PRF = 100 Hz, t_p=100 μs, with predictions from Table IV.

Full downlink slots are predicted at 4.71 Mbps and measured between 4.6–6.8 Mbps, while special slots are predicted at 2.25 Mbps and measured near 3.0 Mbps. The measured average downlink throughput degradation of 8.3% is close to the Case 1 prediction of 7.06%, corresponding to an error of approximately 1.2 percentage points. Across all slot types, the model exhibits a nearly uniform under-prediction offset of approximately 0.6 Mbps. The uniformity of this offset confirms that the collision model is correct and underscores the lower bounds of interference.

B. HARQ Failure Analysis

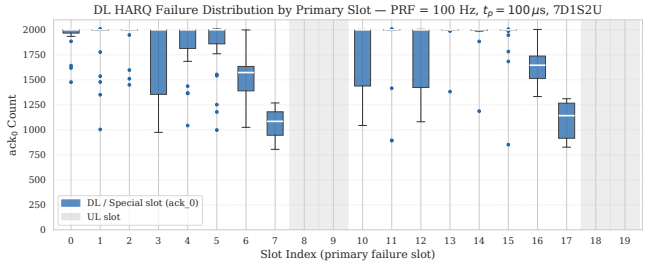


Fig. 6: Interquartile range (IQR) of HARQ failure count distribution for PRF = 100 Hz, t_p = 100 μs, and 7D1S2U.

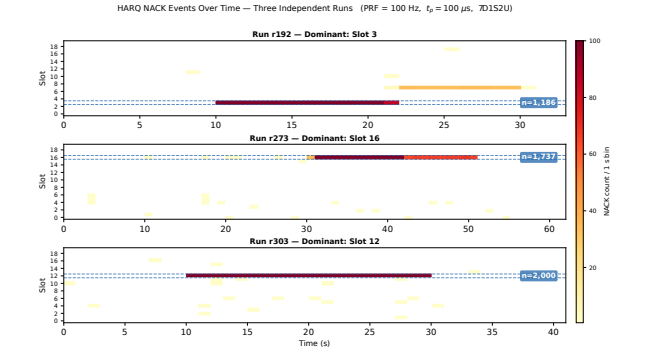


Fig. 7: HARQ NACK events over time for three independent runs at PRF = 100 Hz, t_p = 100 μs, 7D1S2U.

Figure 6 shows the distribution of HARQ failure (NACK) events across identified unique slot failure indices. For a 20 s radar interference event at 100 Hz PRF, we expect roughly 2000 $ack = 0$ events per run for a given slot. Slots 1, 2, 13–15 in Figure 6 show very tight distributions close to this expectation, while slots 3, 10, and 12 has larger IQR variances. Slots 6/7 and 16/17 show a consistent paired reduction. Scheduler log analysis confirms slots 6/16 receive PDSCH schedule grants ∼66% of frames, and 7/17 receive even less ∼58%. (E.g. Fig. 8). This deficit arises from HARQ process exhaustion, where 16 processes are configured and k1 feedback timing is constrained by the UL PUCCH carrying HARQ ACKs, preventing the scheduler from issuing a new transport block until ACKs are returned. Critically, this effect is independent of interference and directly observable in PDSCH grant logs as shown in Figure 8, meaning it can be isolated from collision-induced degradation when fitting the model to live data.

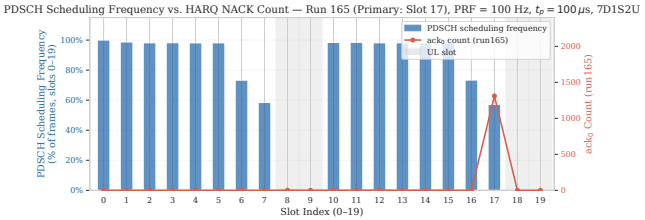


Fig. 8: PDSCH scheduling frequency vs. HARQ NACK count for Run 165.

C. SDR Performance

Figure 7 shows $ack = 0$ event density over time for two outlier runs against a representative clean run (r303), explaining the IQR spread visible in Fig. 6, through SDR/experimental variance. Run r273 shows PRF drift from 100 Hz to 60–70 Hz beginning at $t = 40$ s, indicating a radar-sdr generation fault that reduces the expected NACK count, or scheduler resources as explained in Sec V-B. Run r192 exhibits a timing de-sync at $t = 23$ s followed by a slot shift to slot 6 dominance at 30–40 Hz for the remaining 7 s of the run. Crucially, both failure modes are directly identifiable from the telemetry itself. At the scale of 400+ runs, these artifacts affect individual run counts but leave *dominant identifiable* per-slot averages intact, confirming that SDR hardware imprecision or scheduler resources are separable from true interference behavior without undermining model validation at large run sizes.

D. MultiSlot Failure Analysis

The model predicts that 4% of pulse arrivals (P_3) land within t_p of a slot boundary, straddling consecutive DL slots and producing coupled NACK events across both, as illustrated in Fig. 9. Across over 400 runs, only 9 instances (2.1%) exhibited a measurable secondary DL slot failure, below the prediction of 4%.

The suppression follows from the 2-symbol CORESET ($t_{\text{PDCCH}} = 71.4\,\mu$s), where scheduler logs confirm PDSCH

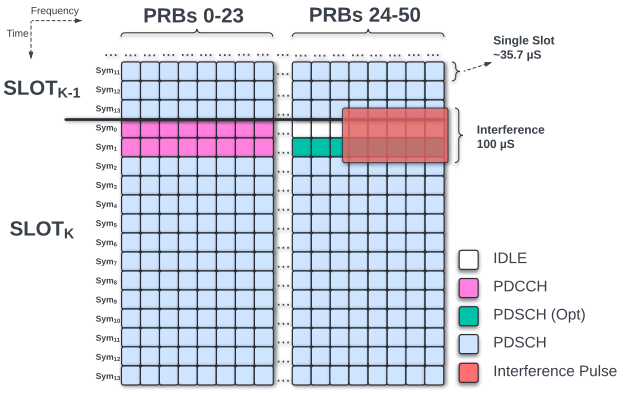


Fig. 9: Physical representation of MultiSlot Case 3 Interference with experimental setup/CORESET configuration from Table V.

occupies only symbols 2–13, so the pulse's overlapping PRBs (45–50) carry no PDCCH for collisions. Cross-slot PDSCH collisions do remain possible when a pulse centered on the blanked CORESET symbols leaves $(100-71.4)/2 = 14.3\,\mu$s of overlap on Sym$_{13}$ of Slot$_{K-1}$ and Sym$_2$ of Slot$_K$. However, this sub-symbol tail produces a NACK only when channel conditions are unfavorable, as reflected in the 9 observed events. In practice, P_3 events rarely produce a multi-slot NACK at this experimental pulse duration, and observed throughput loss is driven almost entirely by single-slot (P_1) collisions.

VI. Conclusion

The OTA results validate the deterministic collision framework across 437 runs to within 1.2 percentage points of measured loss, reproducing the full per-slot lower-bounded degradation model prediction with high accuracy. HARQ failure counts are tightly slot-correlated across runs, which rules out random channel effects and confirms that temporal interference collision is the primary driver of observed degradation. The nearly uniform lower bounded $\sim$0.6 Mbps under-prediction across all slot types points to unmodeled specifics, but also allows for observed variances seen in scheduler behavior. The geometric P_3 term slightly overestimates consecutive-slot failures (2.1% observed vs. 4% predicted), as the 2-symbol CORESET geometry limits boundary-crossing PDSCH exposure to $14.3\,\mu$s per slot side, a sub-symbol tail that produces a NACK only under more specific channel conditions. Separately, HARQ process exhaustion near the UL window reduces PDSCH grant rates for slots 6/7 and 16/17, a scheduler effect due to single UE testing. This scheduling is independent of interference and can be directly separated with MAC KPI/Log processing. Operating without IQ samples or dedicated hardware, the model is well-suited for O-RAN xApp deployments focused on HARQ ID, and Slot specific TBS allocations. The model/experiment validated forward mapping establishes the empirical foundation for Bayesian inverse inference, where MAC-layer anomalies can be used to estimate unknown sensor PRF and pulse duration. Future work will sweep the full PRF range, incorporate HARQ retransmission effects, and extend validation to additional TDD configurations.

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